

Table S1. Technical specifications of the cetacean surveys conducted by the NOAA Pacific Islands Fisheries Science Center included in this analysis.

Survey Year	Study Region	Cruise Number(s)	Hydrophone Type	Hydrophone Frequency Response	Sampling Rate	Total Recording Hours	PAMGuard Version for Detection	Total Encounters for Sensitivity Analysis	Visually-Verified Acoustic Pseudorecords Events	References for further survey information
2010	Hawaiian EEZ	1641; 1642	EDO EC65	2 - 40 kHz	192 kHz	1445	2.02.7b	8; 1	8; 1	Bradford, A. L., Oleson, E. M., Forney, K. A., Moore, J. E., & Barlow, J. (2021). Line-transect abundance estimates of cetaceans in US waters around the Hawaiian Islands in 2002, 2010, and 2017. NOAA Technical Memorandum NMFS-PIFSC-115
2011	Oahu to Palmyra	1108	EDO EC65	2 - 40 kHz	192 kHz	253	2.02.7b	2	2	NA
2012	American Samoa - Palmyra Atoll - Oahu	1203	EDO EC65	2 - 40 kHz	192 kHz	263	2.02.7b	1	1	Pacific Islands Fisheries Science Center (U.S.) Cruise report. (2012). NOAA Ship Oscar Elton Sette, Cruise SE-12-03, 23 April-17 May 2012, Palmyra Atoll Exclusive Economic Zone (EEZ). https://repository.library.noaa.gov/view/noaa/9070 .
2013	Papahānaumokuākea Marine National Monument	1303	APC 42-1021	2 - 40 kHz	192 kHz	293	2.02.7b	2	2	Barkley, Y., Oleson, E. M., Oswald, J. N., & Franklin, E. C. (2019). Whistle classification of sympatric false killer whale populations in Hawaiian waters yields low accuracy rates. <i>Frontiers in Marine Science</i> , 6, 645.
2015	Marianas Archipelago	1502	HTI-96-MIN	2 – 85 kHz	500 kHz	306	2.02.7b	2	2	Hill, M. C., Oleson, E. M., Bradford, A. L., Martien, K. K., Steel, D., & Baker, C. S. (2020). Assessing cetacean populations in the Mariana Archipelago: A summary of data and analyses arising from Pacific Islands Fisheries Science Center surveys from 2010–2019. NOAA Technical Memorandum, (NMFS-PIFSC-108), 98. https://doi.org/10.25923/WRYE-6H14
2016	Main Hawaiian Islands	1604	HTI-96-MIN	2 – 85 kHz	500 kHz	350	2.02.7b	1	1	Barkley, Y., Oleson, E. M., Oswald, J. N., & Franklin, E. C. (2019). Whistle classification of sympatric false killer whale populations in Hawaiian waters yields low accuracy rates. <i>Frontiers in Marine Science</i> , 6, 645.
2017	Hawaiian EEZ	1705; 1706	HTI-96-MIN	2 – 85 kHz	500 kHz	1743	2.00.16	NA	10; 11	Yano, K. M., Oleson, E. M., Keating, J. L., Ballance, L. T., Hill, M. C., Bradford, A. L., Allen, A. N., Joyce, T. W., Moore, J. E., & Henry, A. E. (2018). Cetacean and Seabird Data Collected during the Hawaiian Islands Cetacean and Ecosystem Assessment Survey (HICEAS), July–December 2017. NOAA Technical Memorandum NMFS-PIFSC-72
2018	Marianas Archipelago	1803	HTI-96-MIN	2 – 85 kHz	500 kHz	339	2.01.05	27	0	Hill, M. C., Oleson, E. M., Bradford, A. L., Martien, K. K., Steel, D., & Baker, C. S. (2020). Assessing cetacean populations in the Mariana Archipelago: A summary of data and analyses arising from Pacific Islands Fisheries Science Center surveys from 2010–2019. NOAA Technical Memorandum, (NMFS-PIFSC-108), 98. https://doi.org/10.25923/WRYE-6H14
2019	Main Hawaiian Islands / Papahānaumokuākea Marine National Monument	1901	HTI-96-MIN	2 – 85 kHz	500 kHz	109	2.01.05	10	2	NA
2020	Main Hawaiian Islands	2001	HTI-96-MIN	2 – 85 kHz	500 kHz	475	2.01.05	127	4	Yano, K. M., Oleson, E. M., McCullough, J. L., Hill, M. C., & Henry, A. E. (2020). Cetacean and seabird data collected during the Winter Hawaiian Islands Cetacean and Ecosystem Assessment Survey (Winter HICEAS), January–March 2020. NOAA Technical Memorandum NMFS-PIFSC-111
2021	Marianas Archipelago	2102	HTI-96-MIN	2 – 85 kHz	500 kHz	339	2.01.05	89	10	Yano, K. M., Hill, M. C., Oleson, E. M., McCullough, J. L., & Henry, A. E. (2022). Cetacean and Seabird Data Collected During the Mariana Archipelago Cetacean Survey (MACS), May–July 2021. NOAA Technical Memorandum NMFS-PIFSC-128
2023	Hawaiian EEZ	2303; 2401	HTI-96-MIN	2 – 85 kHz	500 kHz	1168	2.02.15	NA	5	Yano, K. M., Oleson, E. M., McCullough, J. L., Hill, M. C., Bradford, A. L., Joyce, T. W., ... & Henry, A. E. (2025). Cetacean and Seabird Data Collected During the Hawaiian Islands Cetacean and Ecosystem Assessment Survey (HICEAS), July–December 2023. NOAA Technical Memorandum NMFS-PIFSC-175
2024	Southeast of Hawaiian EEZ	2404	HTI-96-MIN	2 – 85 kHz	500 kHz	389	2.02.15	NA	5	NA